

A singularizing random-unitary channel whose Cesaro averages diverge

Counterexample to a mean-ergodic extension suggested in arXiv:2605.24923

Automated mathematics run `fa_banach_001`, lane 5

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Status. Candidate counterexample, likely valid, requiring expert human review. It disproves the universal Cesaro-convergence analogue for singularizing channels and gives an explicit non-quasi-compact example.

1 Source direction and precise scope

Dzhenzher studies random-unitary channels defined by Pettis averaging on possibly nonseparable Hilbert spaces [1]. The conclusion asks for analogues of strong Cesaro convergence for singularizing channels and points to quasi-compactness as a possible route.

$\overline{\Phi}$ itself can be given as the Pettis integral over some measure, which probably should be left-invariant.

6. CONCLUSION

In this paper, we have investigated the structural and dynamical properties of quantum channels Φ_μ and $\mathbf{Q}_{\mu,\pi}$ induced by averaging procedures via the Pettis integral. By establishing a bridge between operator algebra theory and the foundational properties of Ulam real-valued measurable cardinals, we demonstrated how the analytical behaviour of the underlying measure directly dictates the topological nature of the resulting quantum states. Specifically, we provided a complete characterisation of channels that simultaneously preserve σ -additivity while enforcing a strict singularisation of any incoming state. Furthermore, we extended these constructions to general groups of unitary operators, characterising their fixed points and analysing the asymptotic behaviour of their iterations.

The framework developed herein highlights the subtle geometry of singular σ -additive states, which naturally emerge at the intersection of non-separable Hilbert spaces and non-trivial set-theoretic assumptions. While the ergodic properties of normal states under such channels can be successfully resolved via classical mean ergodic theorems, the singularised dynamics pose significantly deeper challenges. As it was mentioned for a separable Hilbert space $\mathcal{H}$ and a σ -additive measure $\mu: 2^{\mathcal{U}(\mathcal{H})} \rightarrow [0, 1]$, the Cesàro averages of $\mathbf{Q}_{\mu,\pi}: \Sigma_n(\mathcal{B}(\mathcal{H})) \rightarrow \Sigma_n(\mathcal{B}(\mathcal{H}))$ converge in SOT to some quantum channel $\Sigma_n(\mathcal{B}(\mathcal{H})) \rightarrow \Sigma_n(\mathcal{B}(\mathcal{H}))$. It is natural to examine the analogous results for singularising quantum channels. However, it is much harder to achieve such results, since we cannot apply the Yosida–Kakutani result (Theorem 5.6). There are analogous results for quasi-compact operators; see, for example, [14, Uniform ergodic theory]. Unfortunately, we have been unable to establish the quasi-compactness of singularising quantum channels $\mathbf{Q}_{\mu,\pi}$ in any particular noteworthy cases.

REFERENCES

Figure 1: The concluding singularizing-channel direction in [1, p. 13].

The source statement is exploratory rather than a numbered yes/no question. We answer its universal reading:

Must the Cesaro averages of a countable-additivity preserving singularizing random-unitary channel converge in strong operator topology? Is such a channel necessarily quasi-compact in the measurable-cardinal setting?

Both answers are negative. The construction uses exactly the source's standing hypothesis that a cardinal carries an atomless countably additive probability measure on all of its subsets.

2 Definitions

For a Hilbert space $\mathcal{H}$, a state is a norm-one positive functional on $\mathcal{B}(\mathcal{H})$. A state is *singular* when it vanishes on $\mathcal{K}(\mathcal{H})$, the compact operators. A channel $\mathcal{Q}$ is *singularizing* when $\mathcal{Q}\rho$ is singular for every state ρ .

For a group representation $\pi : G \rightarrow \mathcal{U}(\mathcal{H})$ and a countably additive probability ν on 2^G , the source's random-unitary channel is

$$\mathcal{Q}_{\nu, \pi} \rho = \int_G \pi(g) \rho \pi(g)^* d\nu(g),$$

where conjugation of a functional is understood by duality. The channel acts linearly and contractively on $\mathcal{B}(\mathcal{H})^*$. Strong operator convergence of channels therefore means pointwise norm convergence on this dual Banach space.

3 Proof intuition

Use two tensor coordinates for two independent jobs. Atomless averaging of translations in the first coordinate singularizes every input: each vector matrix coefficient can see only countably many translations. In the second coordinate, apply one deterministic bilateral shift at every step. After n iterations the state is supported exactly on the n -th second-coordinate slice, regardless of what happened in the first coordinate. Thus the orbit states occupy mutually orthogonal corners. Averaging the first N corners cannot approach averaging the first $2N$: a diagonal sign observable separates the two means by 1 for every N .

4 Construction and proof

Let K be an abelian group of cardinality κ with identity 0. Assume there is a countably additive probability

$$\mu : 2^K \longrightarrow [0, 1], \quad \mu(\{j\}) = 0 \quad (j \in K).$$

This is the Ulam real-valued measurable-cardinal hypothesis used in the source. Put

$$\mathcal{H}_1 = \ell_2(K), \quad \mathcal{H}_2 = \ell_2(\mathbb{Z}), \quad \mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2.$$

Write $(e_k)_{k \in K}$ and $(f_m)_{m \in \mathbb{Z}}$ for the canonical bases. Let $S_j e_k = e_{j+k}$ and let $U f_m = f_{m+1}$.

Take $G = K \times \mathbb{Z}$ and the unitary representation

$$\pi(j, m) = S_j \otimes U^m.$$

Define a probability on all subsets of G by

$$\nu(E) = \mu(\{j \in K : (j, 1) \in E\}).$$

It is countably additive and is supported on $K \times \{1\}$. The corresponding channel is

$$\mathcal{Q}\rho = \int_K (S_j \otimes U) \rho (S_j \otimes U)^* d\mu(j). \tag{1}$$

By the source's preservation theorem, $\mathcal{Q}$ preserves countable additivity [1].

Lemma 1. *The channel $\mathcal{Q}$ is singularizing.*

Proof. Every vector of $\mathcal{H} \cong \ell_2(K \times \mathbb{Z})$ has countable support. Hence, for fixed $x, v \in \mathcal{H}$, the set

$$E_{x,v} = \{j \in K : \langle v, (S_j \otimes U)x \rangle \neq 0\}$$

is countable: any such j is a difference of first coordinates belonging to the two countable supports. Therefore $\mu(E_{x,v}) = 0$.

Let ρ be normal. A positive trace-class operator has a countable spectral decomposition, so $\rho = \sum_r a_r \rho_{x_r}$ with $a_r \geq 0$ and $\sum_r a_r = 1$. For every rank-one projection P_v , positivity and monotone convergence give

$$(\mathcal{Q}\rho)(P_v) = \int_K \sum_r a_r |\langle v, (S_j \otimes U)x_r \rangle|^2 d\mu(j) = 0,$$

because the integrand vanishes outside a countable union of countable sets. Thus $\mathcal{Q}\rho$ vanishes on the compact ideal.

If ρ is singular, then for every compact A and every j , $(S_j \otimes U)^* A (S_j \otimes U)$ is compact, so each integrand in $(\mathcal{Q}\rho)(A)$ is zero. Finally, every state is a convex combination of a normal and a singular state by the normal/singular decomposition recalled in [1]. Hence $\mathcal{Q}\rho$ is singular for every state ρ . $\square$

Theorem 2. *The Cesaro averages*

$$M_N = \frac{1}{N} \sum_{n=1}^N \mathcal{Q}^n$$

do not converge in the strong operator topology on $\mathcal{B}(\mathcal{H})^$. Indeed, for a vector state ρ_0 and every $N \geq 1$,*

$$\|M_N \rho_0 - M_{2N} \rho_0\| \geq 1.$$

Moreover, $\mathcal{Q}$ is not quasi-compact.

Proof. Let $\rho_0 = \rho_{e_0 \otimes f_0}$ and put $\sigma_n = \mathcal{Q}^n \rho_0$. Every unitary appearing in the n -fold integral has second tensor coordinate U^n . Therefore, with

$$R_m = I_{\mathcal{H}_1} \otimes P_{f_m},$$

we have

$$\sigma_n(R_m) = \begin{cases} 1, & m = n, \\ 0, & m \neq n. \end{cases} \quad (2)$$

For $N \geq 1$ define the self-adjoint contraction

$$D_N = \sum_{n=1}^N R_n - \sum_{n=N+1}^{2N} R_n.$$

Equation (2) gives

$$(M_N \rho_0)(D_N) = 1, \quad (M_{2N} \rho_0)(D_N) = 0.$$

Since $\|D_N\| = 1$, it follows that $\|M_N \rho_0 - M_{2N} \rho_0\| \geq 1$. The vector sequence $(M_N \rho_0)$ is not Cauchy, so (M_N) cannot converge strongly.

The same projections show that $\|\sigma_n - \sigma_m\| = 2$ for $n \neq m$: test against $R_n - R_m$. Hence the power orbit of ρ_0 has no relatively norm-compact subsequence. A power-bounded quasi-compact operator has relatively norm-compact power orbits, by its standard finite-dimensional peripheral spectral decomposition [2, Uniform ergodic theory]. Since $\mathcal{Q}$ is a contraction, it is power bounded, and therefore it cannot be quasi-compact. $\square$

5 Verification, scope, and novelty

The construction remains entirely inside the source’s class: ν is a countably additive probability on a group, π is unitary, and Equation (1) is its Pettis random-unitary channel. The large-cardinal measure is not an extra unproved dependency; it is the stated hypothesis under which the source constructs its singularizing channels.

The script `code/check_cesaro_gap.py` verifies with exact rational arithmetic, for $N = 1, \dots, 100$, the scalar weight calculation behind the diagonal witness. It returns gap 1 in every case. This is a finite sanity check, not a substitute for the proof.

A bounded novelty search on 9 August 2026 covered the run registry and indexed web/arXiv searches for the source id and title, singularizing quantum channels, Cesaro nonconvergence, mean ergodicity, and quasi-compactness. It found the source paper and general finite-dimensional quantum ergodicity literature, but no prior result matching this construction or resolving the source’s singularizing direction. Novelty confidence is moderate to high, but the packet should remain a candidate counterexample until expert review.

The result rules out an unrestricted analogue. It does not preclude strong Cesaro convergence or quasi-compactness under additional tightness, compactness, or mixing assumptions.

References

- [1] S. V. Dzhenzher, *Singularizing preserving countable additivity quantum channels on quantum measurable cardinals*, arXiv:2605.24923 (2026).
- [2] N. Dunford and J. T. Schwartz, *Linear Operators, Part I: General Theory*, Wiley-Interscience, 1958.